Supplementary Material S3

Documentary Sources of the Ethical and Clinical Quality Appraisal of the Datasets

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

1. Purpose and procedure

This document supports the table of ethical and clinical quality appraisal of the datasets in the manuscript (Section 2.6), providing the primary documentary source of each judgement. The instrument of eight domains is operationally defined in Section 2.6 of the manuscript.

Two reviewers independently appraised the primary documentation of each dataset (descriptive publication, official repository page, published licence terms or cohort article). Discrepancies were resolved through structured discussion. Where the available documentation derived from works that reused the dataset and not from a primary publication, the domain was judged P (partial) and this is stated accordingly.

In V3.6 the appraisal of ALSPAC was removed, because the only study that used it (Reed et al., 2021) no longer belongs to the eligible corpus, and the SSBD and web video sources used by Kurian and Tripathi (2024) were added. JEMImE remains appraised, as does ABIDE, although the only study that uses it, Dapogny et al. (2019), was moved to contextual evidence in V3.6 by the audit of criterion CI(3). Datasets created by the authors of each study and not publicly released (own clinical recordings) are described in Supplementary Materials S4 and S5 and are not appraised here.

Scale: Y = satisfied with verifiable primary documentation; P = partial or documented only by third parties; N = not satisfied; N/A = not applicable (the dataset does not declare an ASD population); ND = not determinable from the public documentation.

2. Kaggle datasets on childhood autism

Includes Autistic Children Facial Data Set (user imrankhan77), autism-image-data (user cihan063) and Autistic Children Emotions (attributed to F. M. Talaat).

D1 Provenance (N). The original dataset was created by downloading photographs of children from sites dedicated to autism, together with a control set of images collected from the internet. The origin of each individual sample is not traceable: it is not possible to confirm the diagnostic status of any child represented. According to a comment by its creator posted on 10 May 2022, Kaggle withdrew the original dataset for violation of its terms of service.

D2 Consent (N). The retraction notes of PLOS ONE (26 December 2025) state that the consent of legal guardians for the distribution of the images in a public dataset or for their use in academic research is not on record, nor is any ethics committee approval documented.

D3 Diagnostic verification (N). The retraction notes of Scientific Reports indicate that the internet images were collected without documented clinical history or confirmation of an actual ASD diagnosis. The labels are inferred from the web source of the image.

D4 Demographics (N). No metadata on age, sex, ethnicity or geographic origin.

D5 Licence (N). No verifiable declared licence; the original dataset was withdrawn from the platform.

D6 Identification risk (N). Identifiable faces of minors without any de-identification.

D7 Subject separation (N). No subject identifiers.

D8 Clinical applicability (N). None.

Sources: official retraction notes of PLOS ONE (DOI 10.1371/journal.pone.0340338), of Scientific Reports for the studies documented in Section 2.7 of the manuscript, and of Healthcare Technology Letters (*Healthc Technol Lett* 2026;13(1):e70095, DOI 10.1049/htl2.70095) for Ahmad et al. (2024), equally documented in Section 2.7.

3. FER-2013

D1 (P). Created by P.-L. Carrier and A. Courville for the ICML 2013 Workshop on Challenges in Representation Learning. Described in Goodfellow et al. (2013), *Neural Networks* 64:59–63 (2015). Images retrieved through Google Image Search using 184 emotion terms; cropped to 48×48 pixels in greyscale. A descriptive publication exists, but the origin of each individual sample is not traceable.

D2 (N). No consent or ethical approval; collection through search engines.

D3 (N/A). Does not declare an ASD population.

D4 (N). No demographic metadata.

D5 (N). Distributed through the Kaggle competition without a formally declared licence; requires verification per replica.

D6 (N). No formal de-identification.

D7 (N). No subject identifiers.

D8 (N). General non-ASD population; requires domain transfer.

Source: Goodfellow, I. J., et al. (2015). Challenges in representation learning: A report on three machine learning contests. *Neural Networks*, 64, 59–63.

4. FERAC

D1 (P). 770 images of children's faces. The available documentation derives from works that reused the dataset, not from a primary publication. Those works describe it as a modification of a previous dataset hosted on Kaggle, produced in collaboration with clinical personnel from an autism development centre.

D2 (ND). No mention of parental consent or of ethics committee approval appears in the available documentation, which derives entirely from third-party works. The absence of any primary source means the criterion cannot be verified in either direction, which is why it is judged not determinable rather than not satisfied.

D3 (ND). No standardised diagnostic instrument is identified.

D4 (P). The distribution by emotional class is documented, but not the demographic composition.

D5 (ND). Described as an open-access resource, without a specific licence cited.

D6 (N). Images of minors without de-identification.

D7 (ND). Not documented.

D8 (P). Population declared as ASD, but without primary diagnostic verification.

Source: Radočaj, P., & Martinović, G. (2025). *Applied Sciences*, 15(17), 9555, which describes the provenance of the dataset.

5. CK+ (Extended Cohn-Kanade)

D1 (Y). Lucey et al. (2010), *IEEE CVPRW*. 593 sequences from 123 adult subjects aged 18 to 50, collected in a laboratory under an instructed protocol.

D2 (P). Laboratory collection with adult volunteer participants; the details of the consent form are not set out in the publication.

D3 (N/A). Adult population without ASD.

D4 (Y). Documented: 18–50 years, 69% women, 81% Euro-American, 13% Afro-American, 6% other groups.

D5 (Y). Signed use agreement. Restricted to non-commercial research; expressly prohibits providing any portion of the dataset to third parties.

D6 (N). Identifiable adult faces without anonymisation.

D7 (Y). 123 subject identifiers; supports cross-validation leaving one subject out.

D8 (N). Adult population; requires domain transfer.

Sources: Lucey, P., et al. (2010). The Extended Cohn-Kanade Dataset (CK+). *IEEE CVPRW*. Use agreement published by the team responsible for the dataset.

6. CAFE (Child Affective Facial Expression set)

D1 (Y). LoBue & Thrasher, Rutgers University. Validation published in *Frontiers in Psychology* 5:1532 (2015). Children aged 2 to 8 in a laboratory under an imitation paradigm. Deposit documented in Databrary.

D2 (Y). Laboratory collection with minors; distribution requires guardian authorisation and institutional mediation.

D3 (N/A). Typically developing children.

D4 (Y). Documented: 154 children; 90 female and 64 male; 27 Afro-American, 16 Asian, 77 Caucasian, 23 Latino, 11 South Asian. Racial bias relative to the general population declared.

D5 (Y). Access through Databrary; requires registration and authorisation as an accredited researcher.

D6 (P). Images of minors without anonymisation, but restricted access mitigates distribution.

D7 (Y). 154 children identifiable as subjects.

D8 (N). Typically developing population; useful as a control, requires domain transfer.

Sources: LoBue, V., & Thrasher, C. (2015). *Frontiers in Psychology*, 5, 1532. Registration in Databrary, DOI 10.17910/B7301K.

7. DREAM

D1 (Y). Billing et al. (2020), *PLOS ONE*, DOI 10.1371/journal.pone.0236939. EU-FP7 DREAM project. 61 children with ASD; 3,121 robot-assisted therapy sessions and 306 hours recorded at Babeş-Bolyai University, Cluj-Napoca.

D2 (Y). Consent letter signed by at least one parent before the start of the study. Ethical approval from the Scientific Council of Babeş-Bolyai University, registration no. 30664 (10 February 2017). Trial pre-registered at ClinicalTrials.gov, NCT03323931.

D3 (Y). Diagnosis appraised through ADOS, with initial scores between 7 and 20; the ADOS metadata are included in the database.

D4 (Y). Documented: 61 children aged approximately 3 to 6, with age, gender and diagnostic variables.

D5 (Y). Openly accessible through the Swedish National Data Service under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0), DOI 10.5878/17p8-6k13.

D6 (Y). The open publication does not include video recordings or personal information; only three-dimensional processed and anonymised data.

D7 (Y). 61 subjects identified with their respective sessions.

D8 (P). Paediatric ASD population in a therapeutic context, but the open publication contains three-dimensional behavioural data and not facial images. Facial expressiveness monitoring would require access to the unpublished video recordings.

Sources: Billing, E., et al. (2020). The DREAM Dataset. *PLOS ONE*, 15(8), e0236939. Dataset record in the Swedish National Data Service, DOI 10.5878/17p8-6k13.

8. ABIDE (Autism Brain Imaging Data Exchange)

D1 (Y). ABIDE I: Di Martino et al. (2014), *Molecular Psychiatry*, 19(6), 659–667; 17 sites, 1,112 datasets. ABIDE II: Di Martino et al. (2017), *Scientific Data*; 19 sites, 1,114 datasets.

D2 (Y). Each site confirms approval from its local ethics committee both for the initial collection and for the diffusion of the de-identified version. Written informed consent obtained at all sites.

D3 (P). Variable by site: clinical judgement, ADOS or ADI-R, or a combination of both. Diagnostic information is included in the phenotypic data, but the procedure is not homogeneous across sites.

D4 (Y). Documented: age, sex, intelligence quotient and diagnosis. Male predominance declared explicitly.

D5 (Y). Distributed under the protocols of the 1000 Functional Connectomes Project and INDI, in accordance with health privacy regulations.

D6 (Y). Anonymisation with removal of identifiers, including suppression of facial information from the structural images.

D7 (Y). Subject-level identifier; longitudinal data retain the original identifiers.

D8 (N). Neuroimaging, not facial images. Not applicable to facial emotional expressiveness monitoring.

Sources: Di Martino, A., et al. (2014). *Molecular Psychiatry*, 19(6), 659–667. Di Martino, A., et al. (2017). *Scientific Data*, 4, 170010.

9. AffectNet

D1 (P). Mollahosseini, Hasani & Mahoor, *IEEE Transactions on Affective Computing*, 10(1), 2019. More than one million images collected through three search engines with 1,250 emotion terms in six languages. A descriptive publication exists, but the individual origin is not traceable.

D2 (N). No consent; collection through search engines.

D3 (N/A). Does not declare an ASD population.

D4 (P). The original dataset has no demographic metadata; a subsequent extension adds automatically estimated demographic attributes. Class imbalance declared.

D5 (Y). Academic licence agreement; access restricted to principal investigators or laboratory heads.

D6 (N). Identifiable faces without anonymisation; may include unidentified minors.

D7 (N). No subject identifiers.

D8 (N). General population; requires domain transfer.

Source: Mollahosseini, A., Hasani, B., & Mahoor, M. H. (2019). AffectNet. *IEEE Transactions on Affective Computing*, 10(1), 18–31.

10. JEMImE

D1 (Y). ANR CONTINT project no. ANR-13-CORD-0004. Documented in Dapogny et al. (2019), *Frontiers in Computer Science*, and Grossard et al. (2018), *Frontiers in Psychology*, 9, 446. 157 typically developing children aged 6 to 11, recruited in two French public schools (Paris and Nice) between January 2015 and January 2016. Grossard et al. (2018) report 3,875 recorded videos; Dapogny et al. (2019) report 3,768 videos (24 per child) in the version used for their experiments. No children with ASD participated; Grossard et al. (2018) state that the protocol is intended to be proposed to children with ASD in future work.

D2 (Y). Written consent obtained from parents and children before inclusion. Ethics committee approval under reference 15-HPNCL-02.

D3 (N/A). Typically developing children only; no ASD population.

D4 (Y). Documented: 157 children aged 6 to 11 years, 52% male; origin 77.1% Caucasian, 8.3% African, 7% Asian, 7% Maghrebi. Geographic bias between the two centres declared.

D5 (ND). The associated publications are open access, but the access conditions to the video dataset are not publicly declared.

D6 (P). Video of minors' faces without anonymisation; access mediated by the project.

D7 (Y). 157 children identifiable as subjects, with multiple videos each.

D8 (P). Built explicitly to train children to produce facial expressions appropriate to the social context, within a project aimed at children with ASD, and recorded in the paediatric age range of interest. It contains no children with ASD, so applying it to ASD therapeutic monitoring requires domain transfer.

Sources: Dapogny, A., et al. (2019). *Frontiers in Computer Science*, 1, 5. Grossard, C., et al. (2018). *Frontiers in Psychology*, 9, 446.

11. SSBD (Self-Stimulatory Behaviours Dataset) and web videos

Used by Kurian and Tripathi (2024), who combine SSBD with additional videos of children collected from YouTube.

D1 (P). Rajagopalan, Dhall & Goecke (2013), *IEEE ICCV Workshops*, 755–761. 75 videos of about 90 seconds showing three self-stimulatory behaviours (arm flapping, head banging, spinning), collected from videos posted by parents or caregivers on public websites; a later extension of the dataset states that the source websites include YouTube. A descriptive publication exists, but the origin of each individual video is not traceable. The additional YouTube videos of Kurian and Tripathi (2024) have no descriptive publication of their own.

D2 (N). Collection from publicly posted videos; no consent of legal guardians or ethics committee approval for research use is documented in the consulted descriptions.

D3 (ND). No clinical verification of the diagnosis of the children is reported in the consulted descriptions; the labels refer to the behaviour shown in each video.

D4 (ND). No demographic metadata reported in the consulted descriptions.

D5 (ND). No licence is declared in the consulted descriptions; the later extension requires users to accept the licences stated in its repositories.

D6 (N). Identifiable videos of minors without de-identification.

D7 (ND). Not documented.

D8 (P). Paediatric content labelled as autism-related, but annotated for motor behaviours and not for facial emotion, and without verified diagnosis.

Sources: Rajagopalan, S. S., Dhall, A., & Goecke, R. (2013). Self-Stimulatory Behaviours in the Wild for Autism Diagnosis. *Proceedings of the IEEE International Conference on Computer Vision Workshops*, 755–761 (abstract and publisher page; the full text could not be retrieved). SSBD+ extension, arXiv:2311.15072. Appraisal added in V3.6 by one reviewer, pending confirmation by a second reviewer.

12. Generic facial expression datasets on Kaggle

Heterogeneous category of facial expression recognition datasets hosted on Kaggle, distinct from FER-2013 and from the autism datasets. They score N in D1, D2, D4, D6, D7 and D8, and ND in D5, as they typically correspond to republications or aggregations of pre-existing datasets without documentation of provenance, without declared consent, without subject identifiers and with licence labels varying between replicas. D3 is N/A for not declaring an ASD population.

13. Interpretive precisions on selected domain judgements

Three datasets receive judgements whose interpretation is not self-evident from the summary table in the manuscript. They are consolidated here, with cross-reference to the per-dataset entries above.

DREAM, P in D8. The dataset satisfies every other domain with verifiable primary documentation, including diagnostic verification through ADOS and documented parental consent, and it is the most robust public resource identified in this corpus. Its D8 judgement is nonetheless partial because the openly released component contains three-dimensional behavioural data on gaze, head pose and body movement recorded by the robot's sensors, and not facial images.

Applying it to facial expressiveness monitoring would require access to the session video recordings, which are not part of the public release. The judgement therefore reflects a limitation of the released modality, not of the dataset's ethical or clinical quality.

ABIDE, N in D8. The dataset consists of structural and functional magnetic resonance neuroimaging together with phenotypic metadata. It contains no facial images of any kind, and the anatomical scans are subject to defacing precisely to prevent facial reconstruction. It is therefore not reusable for facial expressiveness monitoring under any procedure, which is what the N judgement records. Its remaining domains score favourably; it is included in the appraisal because it is employed by a study documented as contextual evidence (Xu & Ju, 2023), not by a study of the eligible corpus.

FERAC, P or ND across most domains. The pattern of judgements for this dataset differs in kind from the others. Whereas a dataset such as Kaggle ASD scores N because its documentation establishes that the criteria are not met, FERAC scores P or ND because no primary publication of the dataset exists against which any criterion can be verified. Its entire available documentation derives from the works that reused it, which describe it as a modification of a pre-existing Kaggle collection. Parental consent, diagnostic validation and licence terms are consequently not independently verifiable in either direction. This is the reason the manuscript places FERAC in an intermediate provenance category rather than among the datasets with fully verifiable origin, as developed in Section 4.4 of the manuscript.

14. Warning on licence verification

The licence label on the Kaggle page corresponding to FERAC could not be verified through automated consultation and is therefore judged ND; it must be inspected manually before any reuse. The childhood autism emotion datasets are judged N in this domain on different grounds: the original collection was withdrawn from the platform for violation of its terms of service, so no declared licence subsists to be verified. In neither case should a permissive licence be presumed in the absence of an explicit declaration.